UNIVERSITY OF PERADENIYA

FACULTY OF SCIENCE

SEMESTER I - 2024/2025

MAT1073: Mathematics for Operations Research

End Semester Examination

Time allowed: THREE Hours | Answer All Questions

Calculators are not allowed.

Instructions:

Clearly justify each and every step of your solutions with reasoning to receive the assigned number of points.

Notations:

$\mathbb{R}^n$:= Euclidean space of dimension n

$M_{n\times n}(\mathbb{R})$:= set of all $n \times n$ matrices with entries in $\mathbb{R}$

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1. (a) Let $V = \{(x, y) \mid x, y \in \mathbb{R}\}$. Show that the set V forms a vector space over $\mathbb{R}$ under the given addition and scalar multiplication.

$$\text{Addition: } (x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2 - 1, y_1 + y_2 - 1)$$
$$\text{Scalar Multiplication: } c \otimes (x, y) = (cx - c + 1, cy - c + 1), \text{ where } c \in \mathbb{R}.$$

(40 marks)

(b) Consider the following subset in $\mathbb{R}^3$.

$$S = \{(1, 0, 2), (0, 1, 3), (1, 1, 1)\}$$

(i) Show that the set S spans $\mathbb{R}^3$.

(ii) Show that the set S is a basis of $\mathbb{R}^3$.

(iii) Is basis a unique set? Justify your answer.

(40 marks)

(c) Show that $W = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2\times 2}(\mathbb{R}) \;\middle|\; a + b + c = 0 \right\}$ is a subspace of $M_{2\times 2}(\mathbb{R})$. **(20 marks)**

2. (a) Consider the following matrix.

$$A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 0 & 1 \\ -2 & 1 & 1 \end{pmatrix}$$

(i) Write down the **characteristic equation**.

(ii) Find the eigenvalues and the corresponding eigenvectors of A.

(50 marks)

(b) Express the following system of linear equations in the form $A\underline{x} = \underline{b}$ where $A, \underline{x}$ and $\underline{b}$ are matrices in appropriate dimensions.

$$\begin{aligned} 2x + y - z &= 4 \\ x - y + 2z &= 12 \\ 2x + 2y - z &= 9 \end{aligned}$$

(i) Using the Gaussian elimination and back substitution, solve the system of linear equations.

(ii) Is the coefficient matrix A invertible? Justify your answer.

(50 marks)

3. A farm prepares three types of animal feed: cow feed, chicken feed, and goat feed using corn, wheat, and soy. The number of units of each ingredient required for one bag of feed is given below:

Ingredient	**Cow Feed**	**Chicken Feed**	**Goat Feed**
Corn	6	5	6
Wheat	4	6	3
Soy	2	3	5

The farm has 120 units of corn, 100 units of wheat, and 80 units of soy. Let x, y, and z represent the number of bags of cow feed, chicken feed, and goat feed, respectively.

(i) Formulate a system of linear equations to represent the situation.

(ii) Using LU factorization of the coefficient matrix, determine a feasible production plan.

(iii) Later, due to flooding, the farmer loses 30 units of soy. Hence, only 50 units of soy remain available, while the quantities of corn and wheat remain unchanged. Determine whether the reduced soy supply is sufficient for the farmer to produce all three types of feed bags.

(100 marks)

4. (a) Consider the following differential equation:

$$(y \sin x + xy \cos x)dx + (x \sin x + 1)dy = 0.$$

Show that the above differential equation is **exact** and **hence**, solve it. **(20 marks)**

(b) Solve the following Bernoulli differential equation and give the answer in the form $y = f(x)$.

$$2\frac{dy}{dx} + (\tan x)y = \frac{(4x+5)^2}{\cos x}y^3$$

(30 marks)

(c) Using a suitable substitution, solve the following differential equation:

$$\frac{dy}{dx} = \frac{x+2y}{2x-y}.$$

(20 marks)

(d) Find the general solution of each of the following differential equations.

i. $(D^2 - 4D + 3)y = 6x^2 - 2x$

ii. $(D^2 - 2D + 10)y = 10e^x \cos x, \quad$ where $D = \frac{d}{dx}$.

(30 marks)

.END.